

Large Foundation Models and Reasoning

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The question of whether machines can think is like asking whether submarines can swim.

Outline

- 1 Introduction to Reasoning in Large Foundation Models
- 2 Chain-of-Thought Reasoning
- 3 Mathematical Reasoning
- 4 Logical Reasoning
- 5 Advanced Reasoning Techniques
- 6 Reasoning Evaluation
- 7 Future Directions
- 8 Conclusion

What is Reasoning?

Reasoning is the process of drawing logical conclusions from given premises

- **Deductive reasoning:** From general to specific
- **Inductive reasoning:** From specific to general
- **Abductive reasoning:** Best explanation for observations
- **Common sense reasoning:** Everyday logical inference

Key challenge: How can large language models perform complex reasoning tasks?

Types of Reasoning in AI

- ① **Mathematical reasoning**: Solving equations, proofs
- ② **Logical reasoning**: Propositional and first-order logic
- ③ **Causal reasoning**: Understanding cause-effect relationships
- ④ **Commonsense reasoning**: Everyday knowledge and inference
- ⑤ **Multi-step reasoning**: Chain of thought processes

Question: Can foundation models truly reason or just pattern match?

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Chain-of-Thought (CoT) Prompting

Chain-of-Thought prompting encourages models to show their reasoning process

Input $\rightarrow$ Step 1 $\rightarrow$ Step 2 $\rightarrow \dots \rightarrow$ Answer (1)

Example:

- Problem: "Sarah has 12 apples. She gives 3 to Tom and 4 to Mary. How many does she have left?"
- CoT: "Sarah starts with 12 apples. She gives 3 to Tom: $12 - 3 = 9$. She gives 4 to Mary: $9 - 4 = 5$. Sarah has 5 apples left."

Mathematical Formulation of CoT

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be a dataset where:

- x_i : Input problem
- y_i : Ground truth answer

CoT Training Objective:

$$\mathcal{L}_{CoT} = \mathbb{E}_{(x,y) \sim \mathcal{D}} [-\log p(y|x, \text{CoT})] \quad (2)$$

where CoT represents the intermediate reasoning steps.

Key insight: Decompose complex reasoning into simpler steps

Benefits of Chain-of-Thought

- ① **Transparency:** Makes reasoning process visible
- ② **Error localization:** Easier to identify where reasoning fails
- ③ **Improved accuracy:** Especially for multi-step problems
- ④ **Human interpretability:** Aligns with human problem-solving

Empirical evidence: CoT improves performance on:

- Mathematical word problems
- Logical reasoning tasks
- Commonsense reasoning
- Symbolic manipulation

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Mathematical Problem Solving

Large language models can solve mathematical problems through:

- ① **Step-by-step decomposition**
- ② **Algebraic manipulation**
- ③ **Logical inference**
- ④ **Pattern recognition**

Example - Quadratic Equation:

$$x^2 + 5x + 6 = 0 \tag{3}$$

$$(x + 2)(x + 3) = 0 \tag{4}$$

$$x = -2 \text{ or } x = -3 \tag{5}$$

Challenge: Ensuring mathematical correctness in generated solutions

Formal Mathematical Reasoning

For formal mathematical proofs, we need:

$$\text{Given: } \Gamma \vdash \phi \text{ (premises)} \rightarrow \text{Prove: } \psi \text{ (conclusion)} \quad (6)$$

Proof strategies:

- Direct proof: $\Gamma \vdash \phi \rightarrow \psi$
- Proof by contradiction: $\Gamma, \neg\psi \vdash \perp$
- Proof by induction: Base case + inductive step
- Proof by cases: Exhaustive enumeration

Key challenge: Ensuring logical validity of each step

Mathematical Reasoning Capabilities

What LLMs can do:

- Solve arithmetic problems
- Perform algebraic manipulations
- Understand mathematical concepts
- Generate mathematical explanations

Current limitations:

- Inconsistent mathematical accuracy
- Difficulty with complex proofs
- Limited formal verification
- Hallucination of mathematical facts

Research direction: Improving mathematical reasoning reliability

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Propositional Logic Reasoning

Foundation models can perform logical inference using:

$$\text{Premises: } P_1, P_2, \dots, P_n \vdash \text{Conclusion: } Q \quad (7)$$

Logical connectives:

$$\text{Negation: } \neg P \quad (8)$$

$$\text{Conjunction: } P \wedge Q \quad (9)$$

$$\text{Disjunction: } P \vee Q \quad (10)$$

$$\text{Implication: } P \rightarrow Q \quad (11)$$

$$\text{Biconditional: } P \leftrightarrow Q \quad (12)$$

Example: If it rains, then the ground is wet. It's raining. Therefore, the ground is wet.

First-Order Logic Reasoning

For more complex reasoning involving quantifiers:

$$\text{Universal: } \forall x, P(x) \quad (13)$$

$$\text{Existential: } \exists x, P(x) \quad (14)$$

Logical inference rules:

- Modus Ponens: $P \rightarrow Q, P \vdash Q$
- Modus Tollens: $P \rightarrow Q, \neg Q \vdash \neg P$
- Universal Instantiation: $\forall x P(x) \vdash P(c)$
- Existential Generalization: $P(c) \vdash \exists x P(x)$

Challenge: Scaling logical reasoning to complex domains

Commonsense Reasoning

Everyday logical inference that humans perform naturally:

- **Causal reasoning:** Understanding cause-effect relationships
- **Temporal reasoning:** Understanding time sequences
- **Spatial reasoning:** Understanding spatial relationships
- **Social reasoning:** Understanding social interactions

Example: " John went to the store. He bought milk."

- Implicit: John had money, store was open, milk was available
- Inference: John now has milk, spent money, was at store

Key insight: Commonsense requires world knowledge and inference

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Tree of Thoughts (ToT)

Extends CoT by exploring multiple reasoning paths:

Problem $\rightarrow$ Multiple CoT paths $\rightarrow$ Best solution (15)

Process:

- ① Generate multiple reasoning paths
- ② Evaluate each path
- ③ Select the best path
- ④ Refine if necessary

Advantages:

- Explores multiple solution strategies
- Can backtrack from wrong paths
- More robust than single-path reasoning

Self-Consistency

Ensures reasoning consistency by:

Multiple reasoning attempts $\rightarrow$ Consensus answer (16)

Method:

- ① Generate multiple reasoning paths for same problem
- ② Check consistency across paths
- ③ Select most consistent answer
- ④ Flag inconsistencies for review

Benefits:

- Reduces reasoning errors
- Identifies uncertain cases
- Improves overall reliability

Program-Aided Language Models (PAL)

Combines natural language reasoning with code execution:

Problem $\rightarrow$ Generate code $\rightarrow$ Execute $\rightarrow$ Answer (17)

Example:

- Problem: "Calculate the area of a circle with radius 5"
- Generated code: `import math; area = math.pi * 5**2`
- Execution: Returns 78.54
- Answer: "The area is approximately 78.54 square units"

Advantages:

- Precise mathematical computation
- Verifiable results
- Handles complex calculations

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Evaluating Reasoning Capabilities

Key metrics for reasoning evaluation:

- ① **Accuracy:** Correctness of final answers
- ② **Consistency:** Logical coherence of reasoning steps
- ③ **Completeness:** Coverage of all necessary steps
- ④ **Efficiency:** Optimal reasoning path length

Evaluation datasets:

- GSM8K: Grade school math problems
- MATH: High school mathematics
- LogiQA: Logical reasoning
- CommonsenseQA: Commonsense reasoning

Reasoning Error Analysis

Common reasoning errors:

- ① **Logical fallacies:** Invalid inference patterns
- ② **Factual errors:** Incorrect knowledge application
- ③ **Calculation mistakes:** Arithmetic errors
- ④ **Incomplete reasoning:** Missing steps

Error detection methods:

- Consistency checking across multiple attempts
- Step-by-step verification
- External knowledge validation
- Human expert review

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Challenges in Reasoning

Current limitations:

- ① **Hallucination:** Generating plausible but incorrect reasoning
- ② **Inconsistency:** Contradictory reasoning within same response
- ③ **Context limitations:** Difficulty with very long reasoning chains
- ④ **Domain specificity:** Poor generalization across domains

Research directions:

- Improving reasoning reliability
- Scaling to more complex problems
- Better evaluation methodologies
- Integration with external tools

Promising research directions:

- ① **Neuro-symbolic reasoning:** Combining neural and symbolic methods
- ② **Multi-agent reasoning:** Collaborative problem solving
- ③ **Interactive reasoning:** Human-AI reasoning collaboration
- ④ **Formal verification:** Mathematical proof of reasoning correctness

Key questions:

- Can we achieve truly reliable reasoning?
- How to scale reasoning to complex domains?
- What is the role of human oversight?

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Key insights about reasoning in foundation models:

- ① Chain-of-thought prompting significantly improves reasoning
- ② Mathematical and logical reasoning show promising results
- ③ Commonsense reasoning remains challenging
- ④ Evaluation and error detection are crucial

Future outlook:

- Continued improvement in reasoning capabilities
- Better integration with external tools
- More robust evaluation methods
- Potential for truly reliable AI reasoning

Research questions for the community:

- ① How can we ensure reasoning reliability at scale?
- ② What are the fundamental limits of neural reasoning?
- ③ How to best combine symbolic and neural approaches?
- ④ What role should human oversight play?

Contact: For questions and collaboration, please reach out at (hpp@bimsa.cn)

Thank you

Questions and Discussion

Thank you for your attention!